GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS COMPLIANCE:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W3-FLUID-VALVE-NG6_RevA.STEP) GOVERNS NOMINAL GEOMETRY FOR ALL UNTOLERANCED SURFACES.
   1.3 GENERAL TOLERANCES: LINEAR +/- 0.05 mm, ANGULAR +/- 0.2 DEG (PER ISO 2768-m).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS ARE IN MILLIMETERS.

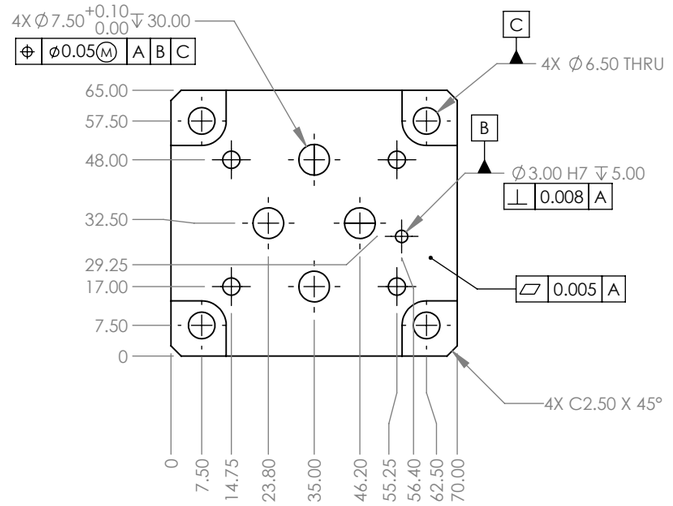
2. MATERIAL SPECIFICATION & HEAT TREATMENT:
   2.1 MATERIAL: HIGH-STRENGTH PRE-STRETCHED AIRCRAFT-GRADE ALUMINUM ALLOY 7075-T651 PER ASTM B209 / AMS 4045.
   2.2 MECHANICAL PROPERTIES (LONGITUDINAL DIRECTION):
      - PROOF STRESS / YIELD STRENGTH (Rp0.2): >= 505 MPa.
      - ULTIMATE TENSILE STRENGTH (Rm): >= 570 MPa.
      - ELASTIC MODULUS: E = 71.7 GPa, DENSITY = 2.81 g/cm^3.
   2.3 STRESS STABILIZATION: CRYOGENIC OR ULTRASONIC RESIDUAL STRESS RELIEF REQUIRED PRIOR TO FINAL PRECISION BORING TO ENSURE DIMENSIONAL INTEGRITY UNDER HIGH PRESSURE.
   2.4 THEORETICAL NET MASS: 560 g (+/- 15 g) (~85.3% REDUCTION VS CONVENTIONAL DUCTILE IRON BLOCKS).

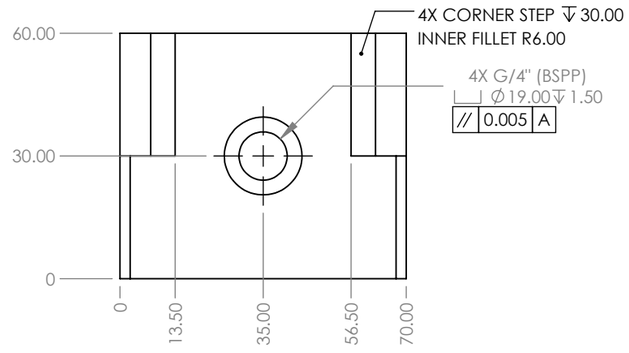
3. PRESSURE RATINGS & FLUID CIRCUITRY INTEGRITY:
   3.1 OPERATING MEDIUM: MINERAL-BASED HYDRAULIC FLUID (ISO VG 32 / VG 46) OR COMPATIBLE DIELECTRIC / SYNTHETIC FLUIDS.
   3.2 WORKING PRESSURE RATING: 31.5 MPa (315 bar) CONTINUOUS SERVICE.
   3.3 STATIC PROOF PRESSURE TEST: 45.0 MPa (450 bar) HELD FOR 5 MINUTES WITH ZERO PRESSURE DROP AND ZERO EXTERNAL WEEPING.
   3.4 MINIMUM BURST PRESSURE DESIGN FACTOR: >= 85.0 MPa (850 bar) (FACTOR OF SAFETY n > 2.7 AGAINST YIELD).
   3.5 ZERO-PLUG MONOLITHIC CIRCUITRY: ALL INTERNAL FLUID PASSAGES (P, T, A, B) ARE STRAIGHT ORTHOGONAL CROSS-DRILLED. NO CONSTRUCTION THREADED PLUGS OR EXPANDER BALLS PERMITTED (ELIMINATES INTERFACIAL HIGH-PRESSURE LEAKAGE RISK).
   3.6 INTERNAL WALL INTEGRITY: STRUCTURAL WALL THICKNESS BETWEEN HIGH-PRESSURE CONDUITS SHALL NOT BE LESS THAN 4.50 mm AT ANY INTERSECTION.

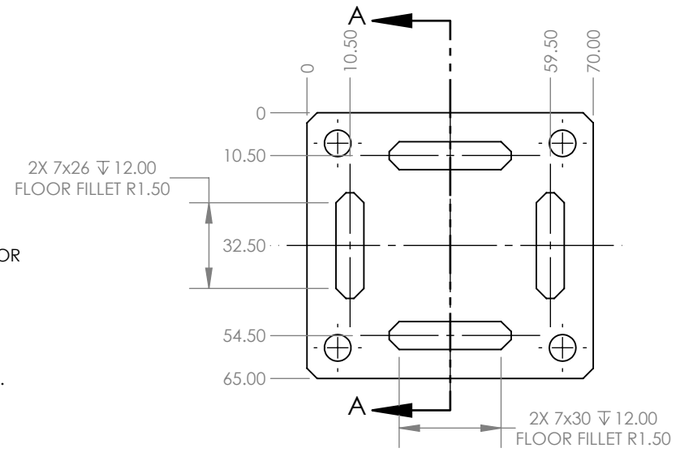
4. PRECISION INTERFACES & GD&T SPECIFICATIONS:
   4.1 TOP MATING INTERFACE (ISO 4401-03-02-0-05 / CETOP 03 / NG6 / NFPA D03):
      - DATUM [A]: TOP VALVE MOUNTING SURFACE. FLATNESS <= 0.005 mm OVER ENTIRE FACE, SURFACE FINISH Ra <= 0.4 MICRONS. ZERO SCRATCHES ACROSS O-RING SEALING CIRCLES PERMITTED.
      - MAIN FLUID PORTS (P, T, A, B): 4X DIA 7.50 +0.10/-0.00 mm, TRUE POSITION [TP] WITHIN 0.05 mm RELATIVE TO DATUM [A] AND LOCATING PIN [B].
      - MOUNTING THREADS: 4X M5X0.8-6H TAPPED HOLES, DEPTH 15.0 mm MIN, ON 40.50 mm X 31.00 mm NOMINAL GRID.
      - ANTI-ROTATION / POLARIZING LOCATING PIN: 1X DIA 3.000 +0.010/-0.000 mm (H7) X 5.00 mm DP LOCATED AT (X=21.40, Y=-3.25) RELATIVE TO VALVE PATTERN ORIGIN.
   4.2 EXTERNAL SERVICE PORTS (LATERAL 4-WAY STAR LAYOUT):
      - 4X G 1/4" BSPP THREADED PORTS (ISO 1179-1 COMPLIANT).
      - SPOTFACE O-RING SEALING FACE: DIA 19.00 +/- 0.10 mm X 1.50 mm DP, SURFACE FINISH Ra <= 0.8 MICRONS.
      - FRONT FACE: P PORT (SYSTEM PRESSURE SUPPLY).
      - REAR FACE: T PORT (TANK / RETURN LINE).
      - LEFT FACE: A WORKPORT (ACTUATOR A-CHAMBER).
      - RIGHT FACE: B WORKPORT (ACTUATOR B-CHAMBER).
   4.3 BASE MOUNTING STEPS: 4X OPEN CORNER STEPPED SHOULDERS AT HALF-HEIGHT (Z=0) WITH 4X DIA 6.50 mm THRU CLEARANCE HOLES TO BOTTOM FOR TOP-DOWN M6 FASTENING.

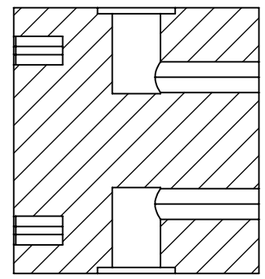
5. CNC DFM MANUFACTURING CONTROLS:
   5.1 CNC MILLING PROTOCOL: 100% SUBTRACTIVE 3/4-AXIS VMC / HMC CONVENTIONAL TOOLPATHS (ZERO LATERAL UNDERCUTS).
   5.2 WEIGHT-REDUCTION POCKETS: 4X PERIPHERAL BOTTOM EDGE POCKETS (DEPTH 12.00 mm, VERTICAL TRANSITION R2.50 mm, FLOOR FILLET R1.50 mm).
   5.3 INTERSECTION DEBURRING: HIGH-PRESSURE THERMAL OR HYDRO-ABRASIVE DEBURRING MANDATORY AT INTERNAL FLUID INTERSECTIONS (P, T, A, B TO G 1/4" JUNCTIONS). 100% ZERO RESIDUAL LOOSE BURRS OR CHIPS PERMITTED.

6. SURFACE TREATMENT & CLEANLINESS SPECIFICATIONS:
   6.1 SURFACE HARDENING: TYPE III HARD ANODIZE COATING PER MIL-A-8625 (CLASS 1, NON-DYED NATURAL).
   6.2 COATING THICKNESS & HARDNESS: 25 TO 30 MICRONS UNIFORM DEPTH; SURFACE MICRO-HARDNESS >= 450 HV FOR WEAR AND CORROSION DEFENSE.
   6.3 CRITICAL MASKING MANDATE: 100% MASKING REQUIRED ON ALL INTERNAL FLUID PASSAGES (DIA 7.50 mm, G 1/4" INTERNALS), TOP O-RING SEALING FACE (DATUM A), AND DIA 3.000 H7 PIN HOLE TO PREVENT COATING ACCRETION AND DIMENSIONAL DRIFT.
   6.4 CLEANROOM CLEANLINESS: COMPLIANT WITH ISO 14644-1 CLASS 3 PARTICULATE STANDARDS.
   6.5 FLUID CLEANLINESS LEVEL: FLUID PATH PARTICULATE PURITY VERIFIED TO ISO 4406 CODE 14/12/9 OR NAS 1638 CLASS 4 VIA MULTI-STAGE ULTRASONIC CAVITATION RINSE IN FILTERED SOLVENT.









SECTION A-A

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH: MIL-A-8625 Type III Hard Anodize (Masked Ports & Datum A)				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION: A
						31.5 MPa (315 bar)	
	NAME	SIGNATURE	DATE			TITLE: HIGH-PRESSURE HIGH-DYNAMIC SERVO VALVE BLOCK (NG6 / CETOP 03)	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: Aluminum Alloy 7075-T651 (ASTM B209 / AMS 4045)		DWG NO. W3-FLUID-VALVE-NG6_RevA	A3
				WEIGHT: 0.56 kg		SCALE: NTS	SHEET 1 OF 1